

Cecil Cao . Jarvis Consulting

I am a Business Systems Analyst at Jarvis, focused on delivering clear insights and actionable solutions to clients. I drive progress using Agile methodologies, and I am most proud of my organizational skills and attentiveness to my team's needs and abilities. I am a CS undergraduate from the University of Toronto, with proven experience as a programming lead in research projects and RA positions investigating new technologies. With 3+ years of professional development, I manage and organize projects and am skilled in full-stack and application development. I am always learning and proactively seek to support my teams. In my free time, I practice digital art, walk in GTA parks, and actively contribute to Toronto's indie game dev community.

Skills

Proficient: C#, Java, Python, Linux/Bash, Agile/Scrum, Git, Typescript

Competent: C++, RDBMS/SQL, Unity, OpenXR, AWS Cloud Computing, Docker

Familiar: GraphQL, Golang, Blender, IsaacSim/IsaacLab, Waterfall

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_CecilCao

Cluster Monitor [GitHub]: Developed a PSQL database hardware usage tracker using Bash scripts and Docker to manage and run the service in a closed network environment. Identified key security risks and reported on the potential risk of SQL Injection attacks.

SQL Database [GitHub]: Investigated a resort dataset containing information about the resort's facilities, reservations, and members. Compiled the data into a relational PSQL database running on Docker and converted high-level questions into SQL queries.

Highlighted Projects

Into the Dataswarm (Beta) [GitHub]: Spearheaded the project as a project manager and programming lead, turning a proof-of-concept game into a fully realized turn-based puzzle game with Trello, Git and Agile Methodologies. Conducted playtesting sessions to discover pain points and bright points, with a post-playtest interview to refine user insights, while making on-the-fly changes on the Unity/C# framework.

Toronto Job Market Database: Researched COVID-19's impact on Toronto's job market through the city's open datasets, converting CSV files into dataframes with Pandas and storing them in a PSQL relational database. Culminated investigation reports into a strategy meeting with the professor, using the number of companies that opened/closed during the pandemic to determine which job sector was heavily impacted, and found that technology sectors generally grew amidst the turbulent time period.

Professional Experiences

Software Developer, Jarvis (2026-present): Interpolate PSQL data from our clients using an ETL pipeline to identify customer loyalty and spending, in order to determine risk and growth factors associated with their business strategy. Used Bash and Docker to run and manage closed network environments that could process and run the database, and Python (Pandas and Pyplot) to clean and visualize the data to present to our customers.

Research Assistant, University of Toronto (2024-2025): Investigated ways to make a large-scale GraphDB RDF graph database more user-friendly with ChatGPT on a RAG framework. Deployed and maintained a Linux cloud service with Python scripts and Docker.

Junior Software Engineer, Alida Inc. (2022-2023): Implemented UX designs on a customer data analytics website using React, Typescript, and a Golang relational database. Organized sprint retros with the team for the team after every bi-weekly Agile sprint to identify key roadblocks.

Education

University of Toronto (2018-2025), Bachelor of Computer Science, Arts and Science

Miscellaneous

- I enjoy hosting creative writing sessions and writing flash fiction pieces for my tabletop RPG campaigns.
- I participate in the International Game Developers Association meetups in Toronto, and am working on an indie game!
- I also work as a private and volunteer tutor on the weekends, teaching K12 students English and Math as my primary subjects.